

BELFAST INTL, United KingdomWMO: **039170**Lat: **54.6638N**Lon: **6.2251W**Elev: **268**StdP: **14.55**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 26.7	(c) 29.7	(d) 22.8	(e) 17.4	(f) 30.6	(g) 25.4	(h) 19.6	(i) 33.0	(j) 31.7	(k) 44.9	(l) 28.8	(m) 46.0	(n) 3.8	(o) 70	(p) 0.637

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.2	(c) 72.7	(d) 62.6	(e) 69.5	(f) 61.5	(g) 67.0	(h) 60.1	(i) 64.5	(j) 69.8	(k) 62.8	(l) 67.6	(m) 61.4	(n) 65.7	(o) 9.0	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.3	84.8	66.6	60.8	80.3	65.1	59.3	76.2	63.6	29.6	70.0	28.5	67.7	27.5	65.7	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 25.6	(b) 22.6	(c) 20.0	(e) 21.3	(f) 77.5	(g) 4.5	(h) 3.2	(i) 18.0	(j) 79.8	(k) 15.4	(l) 81.7	(m) 12.9	(n) 83.5	(o) 9.6	(p) 85.8		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			20.6	67.5	4.3	2.1	17.5	69.0	15.0	70.3	12.6	71.5	9.5	73.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	40.6	41.0	43.1	46.9	51.6	56.4	59.4	59.0	55.7	50.6	44.8	41.3		
	DBStd	8.29	4.96	4.85	4.67	4.68	4.72	4.21	3.92	3.48	3.77	4.57	5.03	5.94		
	HDD50	1406	291	255	217	115	35	2	0	0	3	49	166	273		
	HDD65	5758	755	673	678	543	415	259	179	188	280	446	606	736		
	CDD50	1135	1	2	4	22	85	194	293	280	174	67	10	3		
	CDD65	13	0	0	0	0	0	2	7	4	0	0	0	0		
	CDH74	56	0	0	0	0	1	15	29	11	0	0	0	0		
	CDH80	3	0	0	0	0	0	2	1	0	0	0	0	0		
(14) Wind	WSAvg	9.7	11.0	11.1	10.6	9.9	9.3	8.7	8.2	8.2	8.8	10.0	10.0	10.3		
(15) Precipitation	PrecAvg	33.7	3.2	2.3	2.5	2.2	2.3	2.5	2.8	3.2	3.0	3.4	3.3	3.1		
	PrecMax	42.2	5.9	5.2	4.6	4.6	5.2	6.1	5.7	7.4	6.4	6.9	6.2	7.7		
	PrecMin	27.3	0.9	0.2	0.8	0.6	0.3	1.0	0.6	0.5	0.2	1.2	1.3	0.6		
	PrecStd	3.3	1.2	1.1	0.9	1.1	1.1	1.1	1.1	1.4	1.5	1.5	1.2	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.2	54.9	58.0	64.7	72.1	75.5	77.9	75.2	69.0	62.9	58.1	55.8		
		MCWB	51.8	51.5	51.1	54.8	60.6	61.8	65.6	64.4	62.6	58.7	56.0	53.6		
	2%	DB	51.6	52.4	54.5	60.2	67.2	70.8	73.1	70.2	66.2	60.5	55.8	53.5		
		MCWB	49.8	49.6	50.0	52.7	57.7	60.9	63.1	62.9	60.7	57.3	53.9	51.4		
	5%	DB	50.0	50.5	52.4	57.1	63.3	67.3	69.7	67.3	64.0	59.0	54.0	51.6		
		MCWB	48.4	48.3	48.7	51.2	55.5	59.1	62.0	61.3	58.9	56.1	52.1	49.9		
	10%	DB	48.3	48.6	50.6	54.5	60.0	64.3	66.7	65.2	62.0	57.5	52.4	49.7		
		MCWB	46.6	46.5	47.4	49.5	53.6	57.3	60.2	59.9	57.5	54.8	50.5	48.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	52.2	52.9	56.5	61.9	64.8	67.3	66.7	64.4	60.0	56.6	54.0		
		MCDB	53.5	54.0	56.0	61.2	70.2	72.6	75.1	72.4	67.6	61.5	57.5	55.5		
	2%	WB	50.2	50.2	51.1	54.0	58.7	62.1	64.6	64.1	61.9	58.1	54.3	51.9		
		MCDB	51.3	51.8	53.6	58.9	65.1	68.5	70.7	68.4	65.0	60.0	55.5	53.1		
	5%	WB	48.6	48.5	49.5	51.9	56.7	60.0	62.7	62.5	60.0	56.6	52.5	50.1		
		MCDB	49.8	50.3	51.8	56.1	62.2	65.7	68.2	66.2	62.9	58.6	53.8	51.5		
	10%	WB	46.8	46.8	47.8	50.1	54.6	58.3	61.1	61.0	58.4	55.1	50.8	48.2		
		MCDB	48.2	48.6	50.2	54.0	59.1	63.3	65.7	64.4	61.2	57.2	52.1	49.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	9.4	11.1	12.5	13.6	12.8	12.2	11.6	11.2	9.8	8.7	8.5		
		MCDBR	9.2	10.0	12.4	15.4	18.3	17.9	16.7	14.3	13.1	10.2	9.1	9.9		
		MCWBR	8.9	9.0	9.1	9.7	11.0	10.4	9.5	8.7	8.6	8.2	8.6	9.6		
	5% WB	MCDBR	9.6	9.5	11.2	13.8	17.0	15.9	14.5	12.5	11.4	9.4	8.9	9.9		
		MCWBR	9.4	8.9	9.2	9.5	11.0	10.3	9.2	8.7	8.7	8.4	8.8	9.8		
(40) Clear-Sky Solar Irradiance	taub		0.331	0.342	0.366	0.379	0.384	0.376	0.378	0.362	0.356	0.341	0.325	0.320		
	taud		2.239	2.321	2.295	2.293	2.323	2.374	2.373	2.432	2.446	2.426	2.303	2.204		
	Ebn at Noon		186	233	254	268	273	276	273	271	257	232	189	161		
	Edh at Noon		19	25	32	36	37	36	35	31	27	22	18	16		
(44) All-Sky Solar Radiation	RadAvg		181	396	737	1191	1514	1536	1425	1165	832	470	226	134		
	RadStd		19	36	68	108	135	154	120	94	46	45	24	13		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+1.21	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page